

RESEARCH ARTICLE

Quantifying wildlife watchers' preferences to investigate the overlap between recreational and conservation value of natural areas

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Abstract

1. Nature-based recreation substantially benefits human wellbeing, for example, by improving physical and mental health. However, recreation can also have severe ecological impacts. The recreational value of landscapes and natural areas is often used to generate support for public spending in conservation. However, we still don't know whether nature-based recreationists place greater recreational value on natural areas that have high conservation value compared to other green spaces.
2. Here, we determine which attributes of nature-based tourism provide recreational services. We used pictures of wildlife posted on Flickr to quantify wildlife watching activities in Scotland. We then determined the environmental variables key to attracting wildlife watchers to a destination, such as protected areas (PAs), the perceived naturalness, and the presence of different types of infrastructure.
3. Infrastructure best predicts the intensity of wildlife watching activities in Scotland, while areas of high natural value are rarely used. PAs are weak attractors of wildlife watchers, with PAs designated to protect threatened habitats or species having low recreational value. In accessible and highly visited areas, higher biodiversity increases the intensity of wildlife watching activities.
4. *Synthesis and applications.* Areas of high natural and conservation value and areas of high recreational value do not tend to overlap. Recreational ecosystem services are mainly provided by the wider countryside and highly transformed landscapes, as opposed to wild ecosystems and protected areas designated to protect environmental features of high conservation value. These results question the synergy between the goals of recreation and those of conservation and the use of recreation as a justification for economic investment in conservation. During wildlife watching activities most people experience an urbanised, highly transformed nature; it is important to determine how this human-dominated nature can influence support for conservation of wild and remote areas.

KEYWORDS

conservation, cultural ecosystem services, infrastructure, naturalness, nature-based tourism, protected areas, recreation, urban green networks

1 | INTRODUCTION

Recreation is one of the key cultural ecosystem services offered by nature. It provides substantial benefits for human well-being, including improving physical and mental health (Sandifer, Sutton-Grier, & Ward, 2015). The recreational value of landscapes and natural areas is often used to generate support for public spending in conservation (Balmford et al., 2009; Wilkie & Carpenter, 1999). For example, Protected Areas (PAs) are under increasing pressure to generate income (Walpole, Goodwin, & Ward, 2001). The role of nature-based tourism in conservation is still debated. While tourism can generate economic benefits for local communities and conservation (Krüger, 2005), it faces “leakages”: revenues are often lost from the local area and very little is reinvested into conservation (Sandbrook, 2010). Importantly, it is not clear whether nature-based recreationists are more likely to value natural areas of high conservation value.

All nature-based recreational activities, including hiking and wildlife watching, can have ecological impacts as severe as declines in species richness and shifts in community composition from native to non-native species (Reed & Merenlender, 2008). We need to understand the role of environmental features of high conservation value (e.g., threatened species or high naturalness) in driving attraction of recreationists to a site to determine whether increased recreation might lead to conflicts with conservation goals. Most studies looking at wildlife watchers' and nature recreationists' preferences have focused on visitors to PAs (Baum, Cumming, & De Vos, 2017; De Vos, Cumming, Moore, Maciejewski, & Duckworth, 2016; Sessions, Wood, Rabotyagov, & Fisher, 2016; Sonter et al., 2016), thus missing an important part of nature recreationists that use green spaces in urban areas or the wider countryside. Our view on preferences is therefore likely biased towards specialists, who will have different preferences from the general public. Also, they might have underestimated the recreational value of some environmental features or infrastructure that is not represented in PAs.

Here we identify the natural and human characteristics of a destination that predict high intensity of recreational activities. We focused on wildlife watching in Scotland, where nature-based tourism contributes £1.4 billion per year to the economy, creating 39,000 Full Time Equivalent jobs, with £153 million attributable to wildlife watching alone (Bryden, Westbrook, Burns, Taylor, & Anderson, 2010). Surveys are the most common tools to quantify recreational activities (Casado-Arzuaga, Onaindia, Madariaga, & Verburg, 2014; Peña, Casado-Arzuaga, & Onaindia, 2015) and questionnaires are mostly used to understand visitors' preferences (van Zanten, Verburg, Scholte, & Tieskens, 2016). The widespread use of social media offers now a great opportunity to use crowdsourced datasets, allowing us to sample more people. A few studies have compared geotagged photographs uploaded on social media to visitor statistics obtained through more traditional methods (e.g., surveys or censuses) (Hausmann, Toivonen, Slotow, et al., 2017; Levin, Kark, & Crandall, 2015; Levin, Mark, & Brown, 2017; Sessions et al., 2016; Wood, Guerry, Silver, & Lacayo, 2013), and all of these studies have demonstrated that data from social media are a reliable indicator

of intensity of recreational activities and preferences. We used the number of users of the photo-sharing website Flickr that posted photographs of wildlife taken in Scotland as our measure of intensity of wildlife watching activities (Mancini, Coghill, & Lusseau, 2018). We assume, as in previous studies, that the more people photograph and share information about a particular location, the higher its recreational value and we interpret spatial concentration of social media content as an indicator of the popularity of a wildlife watching destination (Gliozzo, Pettorelli, & Haklay, 2016; van Zanten, van Berkel, et al., 2016).

The potential of ecosystems to provide recreational services depends on different factors, such as their beauty, naturalness and biodiversity, and the presence of a PA, but accessibility and infrastructure are also crucial (Maes et al., 2011; Peña et al., 2015). For example, large areas of Scotland have semi-natural landscapes that show minimal signs of current human influence. These can be a variety of habitat types, from mountains to undeveloped coastline, but they are all characterised by high naturalness and biodiversity (therefore high conservation value), low accessibility and no infrastructure. Areas that have higher naturalness and biodiversity, protected under certain designations (Site of Special Scientific Interest or Marine PAs) will also have a high conservation value because they were designated to protect threatened species or important habitats, but they will provide better accessibility and more infrastructure compared to wild areas. Other PA designations, such as Country Parks (CNTRY) or Local Nature Reserves (LNR) state as their main goal to provide opportunities for the public to enjoy nature close to where they live and were designated with the dual objective of preserving important natural and cultural heritage and providing people with recreational opportunities. Therefore, these areas will have lower conservation value compared to PAs such as Marine PAs and to wild areas. Following this conceptual framework, we tested the effect of different types of PA designations, naturalness, biodiversity, accessibility, and presence of recreational infrastructure on the intensity of wildlife watching activities in Scotland. We aimed to assess whether wildlife watchers place greater recreational value on natural areas that have high conservation value compared to other green spaces.

2 | MATERIALS AND METHODS

2.1 | Data collection

2.1.1 | Wildlife watching

To quantify wildlife watching activities in Scotland, we queried the Flickr Application Programming Interface (API) for photographs of wildlife taken in Scotland between 2005 and 2015. We used four keywords to select only relevant photographs: “bird,” “seal,” “whale,” and “dolphin,” which are the main groups of charismatic wildlife watched in Scotland (Curtin, 2013). We used packages RCurl (Lang & The CRAN Team, 2015a), XML (Lang & The CRAN Team, 2015b) and httr (Wickham, 2016) in R (R Core Team, 2015) to communicate

with the API, and request and download the data. We downloaded the metadata associated with the photos: photograph and user ID, the date when the photo was taken, the geographic coordinates of where it was taken, user tags and description. Using the user tags associated with the photos, we eliminated pictures that were not relevant (such as photos of statues or paintings and photos taken in zoos). The tags were examined and a list of keywords for non-relevant photos was compiled, then, following a method similar to that used in (van Zanten, Van Berkel, et al., 2016), we used that list of keywords to filter out irrelevant photos. In order to avoid bias coming from having a small number of very active users, we used the combination of user ID and date to delete multiple photos from the same user on the same day. Therefore, the number of data points in the dataset (41203) represents the number of Flickr visitor days (FVD; Figure 1). For details on this data collection procedure see (Mancini et al., 2018 and Data Sources)

2.1.2 | Infrastructure

Information about infrastructure was downloaded from Google Places API Web Service and Ordnance Survey (see Supporting Information Table S1 and Data sources). Google Places API was queried to obtain the locations of tourist accommodations, tour operators offering wildlife recreational activities, airports, bus stops, train stations, and car parks (Figure 1). We used the packages *httr* (Wickham, 2016) to query the API and *jsonlite* (Ooms, 2014) to format the data, and we used the Google Places API Radar Search Services to perform the search for the different types of infrastructure on the Google Places API. Because of the limits in the search results returned by this service (maximum 200 places), we created a grid of 2×2 km cells over Scotland and used the coordinates of the cell centres for the search, with a radius of 2.5 km. This search produced duplicates, which were then removed using the place ID. We used the “type” argument in the Google Places API Radar Search Services to download information on the different types of infrastructure: “lodging,” “airport,” “bus_station,” “train_station,” and “parking.” We further filtered the airport list by only selecting those with airline services according to OurAirports, a website that provides a collection of aviation data for airports around the world. (<http://ourairports.com/countries/GB/SCT/airports.html?show=scheduled>). For tour operators, we used both “type=tour_operator” and “keyword=wildlife” to restrict the search to wildlife-related tours only. This search returned some non-relevant results, therefore we went through each location returned by the API and manually selected only those operations that charged money for some kind of wildlife watching infrastructure, such as boat tours, guided walks, visitor centres with cameras on bird nests or bird hides. Each of these searches only returned the place ID, we then used these IDs in Place Details Requests to obtain the coordinates and name of each infrastructure. In total we obtained the locations of 14057 tourist accommodations, 22 airports, 41,166 bus stops, 372 train stations, 693 car parks, and 112 tour operators.

We also downloaded a shapefile of roads from Ordnance Survey Open Data (Supporting Information Table S1, Data sources and

Figure 1) to test the effect of the presence of roads on intensity of wildlife watching.

2.1.3 | Environmental data

All the environmental variables tested were obtained from Scottish Natural Heritage (SNH) Natural Spaces application (Supporting Information Table S1 and Data Sources), except the biodiversity records which were downloaded from the Global Biodiversity Information Facility (GBIF, Supporting Information Table S1 and Data sources).

From SNH Natural Spaces we downloaded shapefiles with boundaries for PAs in Scotland (Figure 1). These areas include: local designations, such as LNR, Marine Consultation Areas, SNH Nature Reserves (NR) and Country Parks (CNTRY); national designations, such as Sites of Special Scientific Interest (SSSI), National Nature Reserves (NNR), National Parks (NP) and Marine Protected Areas (MPA); international designations such as Council of Europe Diploma Sites (COUNEUR), Natura Sites (Special Areas of Conservation—SAC—and Special Protection Areas—SPA), Ramsar sites, Biosphere (BIOSPH) and Biogenetic (BIOGEN) reserves and World Heritage Sites (WHS).

From SNH Natural Spaces we also downloaded a raster of perceived naturalness in Scotland at a resolution of 1 m (Figure 1). This naturalness score was estimated previously from a land use map, where each land class was given a naturalness score from 1 (*low naturalness*) to 5 (*high naturalness*); then a focal statistical window of 250 m was passed over the dataset averaging the naturalness value to account for surrounding areas (for more details see Carver et al., 2008).

Records of species occurrence were downloaded from GBIF (Supporting Information Table S1 and Data sources). We queried the GBIF database for occurrences of Phocidae, Aves and Cetacea in Scotland, which returned 333,105 occurrences (Figure 1).

2.2 | Data manipulation

We showed previously that FVD counts is an unbiased measure of wildlife watching intensity at a resolution as low as 10×10 km (Mancini et al., 2018). Therefore, we aggregated all variables previously described at this resolution. Data were processed in ArcGIS 10.4.1 (Supporting Information Table S2).

The response variable, intensity of wildlife watching (Count_WW), was obtained by counting the number of FVD in each 10×10 km cell (Figure 1). Because the naturalness score was only available for terrestrial locations, we excluded all the photos that were taken at sea (135 photographs). The polygon shapefiles representing the boundaries of the different types of PAs were processed by calculating the area of each cell that was covered by that type of PA (Area_*). For marine PAs we calculated the distance between the centres of each cell to the boundaries of the nearest reserve instead (Dist_*). We also created an aggregated variable, where we combined all the types of terrestrial PAs into one layer and then calculated for

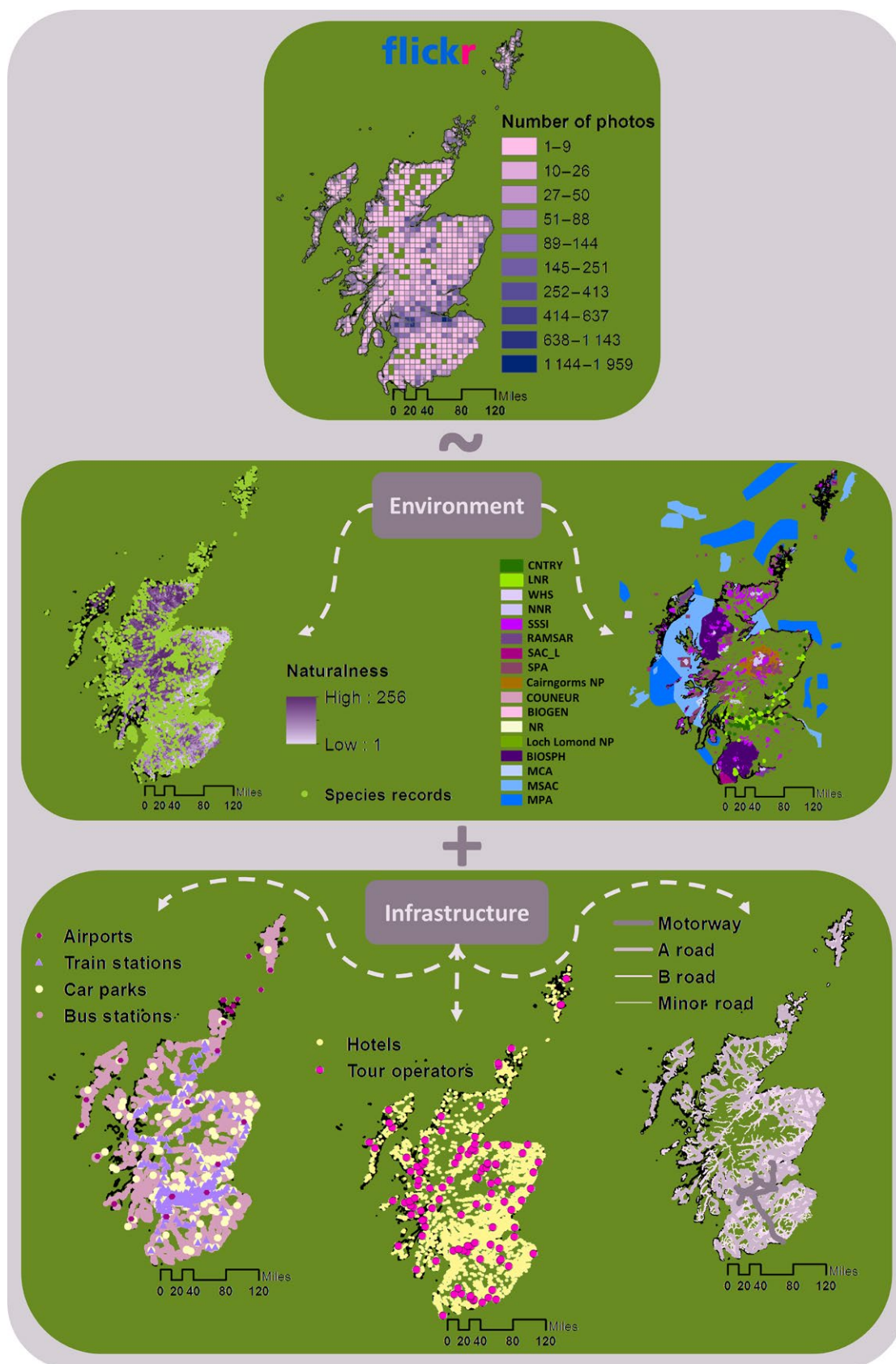


FIGURE 1 Framework and data used in the study. The response variable is the number of Flickr users taking pictures of wildlife in each 10×10 km grid cell (top box). We tested the effect of different environmental (middle box) and infrastructure (bottom box) variables on the response variable. Some of the offshore marine PAs are not displayed on the map because of space limits. All the maps were created in ArcGIS 10.5. For data sources see text and Supporting Information Table S1

each cell the area covered by any type of PA (Area_PA). For each cell we also calculated an average naturalness value by calculating the mean of the naturalness score across all the raster cells within our 10×10 km grid cells (Mean_Nat). We then calculated the number of and the distance from the centres of each cell to the nearest airport (Dist_Air), car park (Dist_CarPark), tour operator (Dist_TourOp), train station (Dist_Train) and road (Dist_Road), and the number of tourist accommodations (Count_Hotels) and bus stops (Count_Bus) in each cell. As we did for PAs, we calculated an aggregated infrastructure variable which consisted of the count of any type of infrastructure inside each cell (Count_Inf). Species richness was calculated by counting the number of unique species recorded in the GBIF data in each cell (Species). For occurrences of marine species, we only selected those that were within 1 km from the coast, we then assigned each occurrence to their nearest coastal grid cell and added these to the number of terrestrial species occurrences in the same cell.

2.3 | Analysis

The aim of this analysis was to understand how environmental and infrastructure variables explained the variance in the intensity of wildlife watching at a 10×10 km resolution. First, we wanted to know which aspect of a destination (e.g., the infrastructure, the presence of PAs or its perceived naturalness) is most important in explaining intensity of wildlife watching. Then we tested the effect of each specific infrastructure, type of PA, or environmental variable (naturalness or species richness). Therefore, we performed

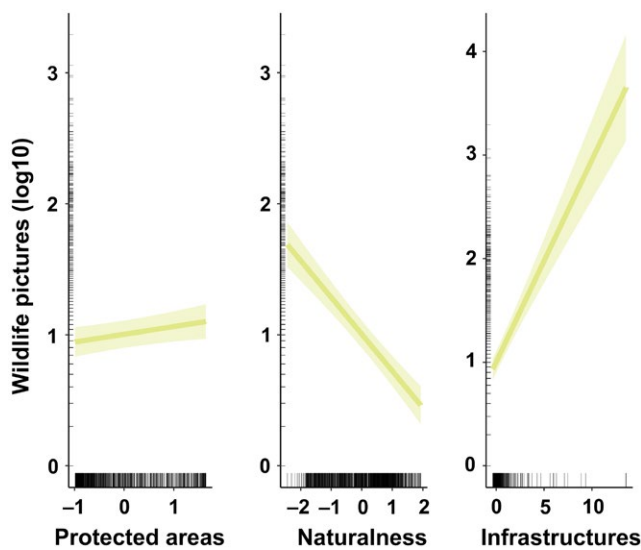


FIGURE 2 Results of the aggregated model. The plots show predicted values of Flickr users photographing wildlife against the area covered by a protected area (left), the mean perceived naturalness (centre) and the number of infrastructures (right) in the cell while other variables are kept at their mean. All the variables on the x-axis were centred around the mean and scaled by their SD. Ticks on the axes represent the data, the lines represent the predictions from the model and the lighter shaded areas are the 95% confidence intervals

our model selection in three steps. We first regressed our response variable, the intensity of wildlife watching against aggregated variables (area covered by a PA, number of infrastructure) and mean naturalness score (Supporting Information Table S1). We then fitted an infrastructure model, where we estimated the effect of each type of infrastructure on the response variable, and an environmental model where we tested the effect of each different type of PA and of the mean naturalness score on the intensity of wildlife watching. We adopted this approach to avoid overfitting due to the high number of covariates that we would have had to use in one full model. For the same reason and due to the limited sample size (1,132 observations), we were not able to test for the effect of interactions between some of the explanatory variables, which would have resulted in low degrees of freedom and over fitted models. Moreover, given the degree of correlation between some of our explanatory variables, for example different types of infrastructure, we didn't fit interaction terms to avoid over inflation. We also performed an analysis in which all the predictors selected in the infrastructure and environmental models were tested together to directly compare their effects on the intensity of wildlife watching (Supporting Information Table S3).

The spatial distribution of species records from GBIF overlapped the distribution of FVD. Some areas of Scotland are difficult to access, the terrain is challenging and generally are only visited by experienced hikers. In these areas both the intensity of wildlife watching and the number of GBIF records were zero or very low (Supporting Information Figure S1 in Appendix S1). As a consequence of this, we decided to subset the dataset according to the number of GBIF records so we only considered those areas where enough data were present. The distribution of this variable was bimodal, with one mode around 10 and one around 10,000 records (Supporting Information Figure S2 in Appendix S1). We fitted a mixture model to the number of GBIF records to obtain a classification of our observations using the function `densityMclust` in the R package `mclust` (Fraley & Raftery, 2002; Fraley, Raftery, Murphy, & Scrucca, 2012) to classify our observations into two groups, high and low GBIF records. We then only used the observations belonging to the "high" group to test the effect of the number of species on the intensity of wildlife watching.

Given the skewness of the response variable we $\log_{10}$ transformed it. We checked for collinearity between our explanatory variables by estimating variance inflation factors (VIF). All the variables had VIF <2 except the area covered by a Site of Special Scientific Interest and area covered by a terrestrial Special Area of Conservation for which VIF was >3; therefore, these two variables were not used together in the same models.

We first fitted a linear model using $\log_{10}$ -transformed intensity of wildlife watching as response variable and all the other variables as explanatory variables. Inspection of the residuals from this model revealed the presence of spatial autocorrelation. In order to account for these spatial patterns we used linear regressions fitted with Generalised Least Squares, which allows for the inclusion of autocorrelation structures. We used the function `gls` in the R package `nlme`

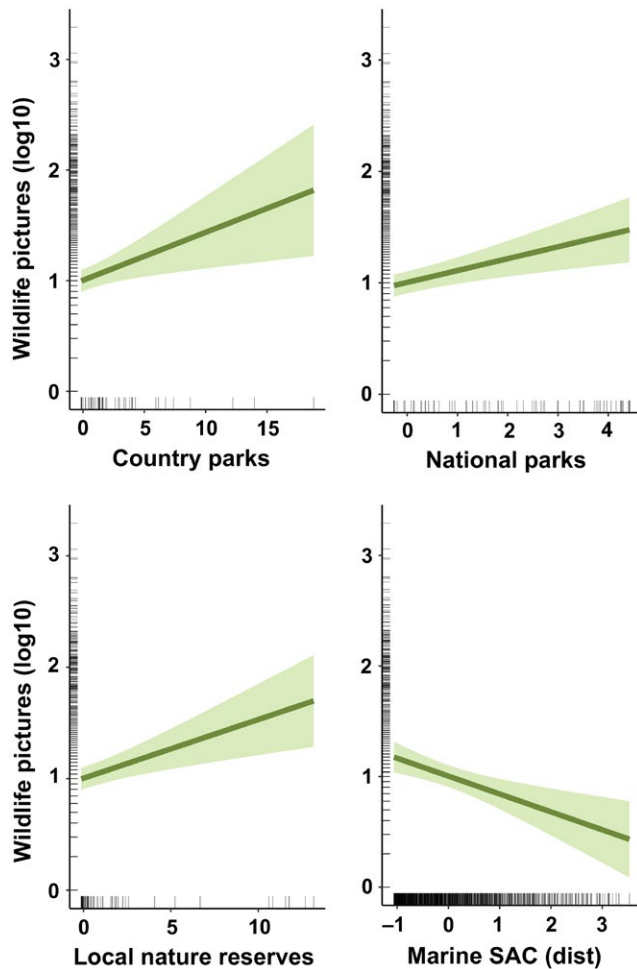


FIGURE 3 Results of the environmental model. The plots show predicted values of Flickr users photographing wildlife against the area covered by a Country Park (top-left), a National Park (top-right), a Local Nature Reserve (bottom-left) and the distance from a marine Special Area of Conservation (bottom-right) while other variables are kept at their mean. All the variables on the x-axis were centred around the mean and scaled by their SD. Ticks on the axes represent the data, the lines represent the averaged predictions from the best models and the lighter shaded areas are the 95% confidence intervals

(Pinheiro, Bates, Debroy, & Sarkar, 2016). All the covariates were centred around the mean and scaled by their SD so that coefficients were comparable.

2.3.1 | Aggregated model

The first model we fitted is described in Equation 1:

$$\log_{10}(\text{Count_WW}) \sim \text{Area_PA} + \text{Count_Inf} + \text{Mean_Nat} \quad (1)$$

We fitted five models with different correlation structures (rational quadratic, spherical, linear, Gaussian and exponential) with restricted maximum likelihood (REML) and then used AIC to select the retained correlation structure. We then used diagnostic plots and variograms of the model residuals to check that model assumptions were not violated.

2.3.2 | Environmental model

This model is described by Equation 2:

$$\begin{aligned} \log_{10}(\text{Count_WW}) \sim & \text{Area_WHS} + \text{Area_SAC_L} + \text{Area_SSSI} \\ & + \text{Dist_MSAC} + \text{Dist_MPA} + \text{Dist_MCA} + \text{Area_SPA} \\ & + \text{Area_RAMSA} + \text{Area_NR} + \text{Area_NNR} + \text{Area_LNR} \\ & + \text{Area_COUNE} + \text{Area_CNTRY} + \text{Area_BIOSP} + \text{Area_BIOGE} \\ & + \text{Area_NP} + \text{Mean_Nat} \end{aligned} \quad (2)$$

After choosing the best autocorrelation structure using the same method we used for the aggregated model, we performed model validation by visually inspecting diagnostic plots and variograms of model residuals, then we performed model selection. Given the high number of variables we decided to use a data mining approach. We used the function dredge in the R package MuMIn (Barton, 2016) to fit models with all the possible combinations of the explanatory variables (without interactions), except for models that contained the two collinear variables together. All the models were fitted with maximum likelihood and after AIC and weights were calculated we refitted all the models with a $\Delta\text{AIC} < 5$ with REML to obtain unbiased coefficients. We then calculated averaged coefficients across the best ($\Delta\text{AIC} < 5$) models.

2.3.3 | Infrastructure model

In this model we tested the effect of the different types of infrastructure:

$$\begin{aligned} \log_{10}(\text{Count_WW}) \sim & \text{Dist_Air} + \text{Count_Bus} + \log_{10}(\text{Count_Hotel}) + \\ & \text{Dist_CarPark} + \text{Dist_TourOp} + \text{Dist_Train} + \text{Dist_Road} \end{aligned} \quad (3)$$

Model validation and selection were performed as for the environmental model.

2.3.4 | Species richness model

After sub setting the dataset according to the number of GBIF records (mixing probabilities: low 0.13, high 0.87; means: low 18, high 25118), we used the variables that were selected as important by the model selection procedure on the environmental model and used them as covariates together with the number of species in this model:

$$\begin{aligned} \log_{10}(\text{Count_WW}) \sim & \text{Dist_MSAC} + \text{Area_LNR} + \text{Area_CNTRY} + \\ & \text{Area_NP} + \text{Mean_Nat} + \text{Species} \end{aligned} \quad (4)$$

For every model we then produced maps of model residuals (averaged for the environmental and infrastructure models) to investigate how well our models fit the data and identify any spatial patterns left unexplained.

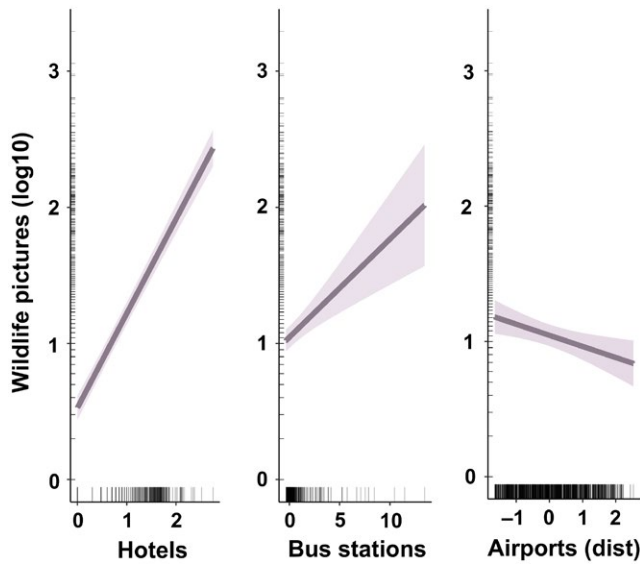


FIGURE 4 Results of infrastructure model. The plots show predicted values of Flickr users photographing wildlife against the number of tourist accommodations (left), the number of bus stops (centre) and the distance from the nearest airport (right) while other variables are kept at their mean. All the variables on the x-axis were centred around the mean and scaled by their SD. Ticks on the axes represent the data, the lines represent the averaged predictions from the best models and the lighter shaded areas are the 95% confidence intervals

3 | RESULTS

We retained an exponential correlation structure for all models.

3.1 | Aggregated model

All the variables had a significant effect on the intensity of wildlife watching (Figure 2). The amount of infrastructure present in each 10×10 km cell had a positive effect on the response variable, the intensity of wildlife watching (Figure 2 right; coefficient: 0.2, SE: 0.02, t -value: 10.4, df : 1,131, $p < 0.001$). The area of the cell that is occupied by a protected area also had a positive, but weaker, effect on the response variable (Figure 2 left; coefficient: 0.06, SE: 0.02, t -value: 2.45, df : 1,131, $p < 0.01$), while the mean naturalness score of the area had a negative effect (Figure 2 middle; coefficient: -0.3 , SE: 0.03, t -value: -10.12 , df : 1,131, $p < 0.001$).

3.2 | Environmental model

Wildlife watching intensity increased as the distance from a marine Special Area of Conservation (coefficient: -0.16 , adjusted SE: 0.05, z : 3.4) and the area's mean naturalness score (coefficient: -0.4 , adjusted SE: 0.03, z : 13.8) decreased and as the area covered by a Country Park (coefficient: 0.04, adjusted SE: 0.01, z : 2.7), the area covered by a Local Nature Reserve (coefficient: 0.05, adjusted SE: 0.01, z : 3.4) and the area covered by a National Park (coefficient: 0.1, adjusted SE: 0.03, z : 3.4) increased (Figure 3).

3.3 | Infrastructure model

Number of tourist accommodations (coefficient: 0.4, adjusted SE: 0.01, z : 25.3) and number of bus stations (coefficient: 0.07, adjusted SE: 0.01, z : 4.3) increased the intensity of wildlife watching in each 10×10 km cell (Figure 4), while distance from an airport had a negative effect (Figure 4; coefficient: -0.08 , adjusted SE: 0.03, z : 2.7).

3.4 | Species richness model

In areas where there was sufficient access for people to visit, the strongest positive effect on the intensity of wildlife watching was that of the number of species (Figure 5; coefficient: 0.5, SE: 0.04, t -value: 13.36, df : 760, $p < 0.001$). The other variables maintained similar effects as in the previous environmental model: distance from marine Special Areas of Conservation (coefficient: -0.08 , SE: 0.03, t -value: -2.35 , df : 760, $p < 0.05$), area covered by a Local nature Reserve (coefficient: 0.03, SE: 0.01, t -value: 2.01, df : 760, $p < 0.05$), area covered by a Country Park (coefficient: 0.04, SE: 0.02, t -value: 2.64, df : 760, $p < 0.05$), area covered by a National Park (coefficient: 0.08, SE: 0.03, t -value: 2.63, df : 760, $p < 0.05$), and mean naturalness score (coefficient: -0.23 , SE: 0.02, t -value: -8.58 , df : 760, $p < 0.001$).

The maps of model residuals (Figure 6) showed some spatial patterns left unexplained by the models. The aggregated and environmental model (Figure 6 top) seemed to be under-predicting the intensity of wildlife watching in the more populated part of Scotland, the central belt of Scotland, especially around Glasgow and Edinburgh, in the area around Inverness and the Cairngorms National Park, on the west coast and in North and South Shetland. The same models were over-predicting the intensity of wildlife watching in the Highland and in the southern regions of Ayrshire and Dumfries and Galloway. The residuals from the species richness model seemed to present a very similar pattern (Figure 6 bottom-left). The infrastructure model provided the best fit to the data, with less over- and under-prediction compared to the other ones (Figure 6 bottom-right).

4 | DISCUSSION

Interactions with wildlife were common outside PAs which have been the primary focus of studies on recreational ecosystem services so far (Balmford et al., 2009; Baum et al., 2017; De Vos et al., 2016; Sessions et al., 2016; Sonter et al., 2016). Secondly, wildlife watchers in Scotland experience wildlife mostly in areas that have low conservation value, where nature is easily accessible and facilities are provided, while areas that have very high naturalness are rarely used (Figures 2, 4, and 6). The preferred PAs, Country Parks, LNR and NP (Figure 3), are all areas that provide opportunities for the public to enjoy nature close to where they live (Supporting Information Figure S3 in Appendix S1). They were designated with the dual objective of preserving important natural and cultural heritage and providing people with recreational opportunities. As such, they are better connected to infrastructure to facilitate access.

NNR, SSSI or Special Protection Areas, which are managed for the conservation of threatened species or important habitats, have limited infrastructure in their vicinity because of their remit. They are also not intensely used by wildlife watchers, except for marine Special Areas of Conservation, where we found a higher intensity of wildlife watching (Figure 3). We think this is because coastal Special Areas of Conservation are very close to main cities in the East (e.g., Inverness, Dundee, and Edinburgh) and to very popular tourist areas in the West (Supporting Information Figure S4 in Appendix S1).

The nature that the people experience during wildlife watching activities is not the one on which conservation programmes are focussed; it is a nature with a strong human dimension. In the context of wildlife watching, wild landscapes and environmental features that are priorities for conservation do not attract recreationists as other types of low-conservation value green space do. Hence, it is important to rethink the position of green spaces in providing recreational ecosystem services. The concept that high conservation value does not enhance recreational ecosystem services was proposed by a recent study (Hornigold, Lake, & Dolman, 2016) where the authors compared visitation to similar natural areas inside or outside SSSI as a proxy for the effect of high conservation value on the likelihood of site visitation. Our study expands on this work by directly comparing the effect of

infrastructure versus that of naturalness on the intensity of wildlife watching and by testing the effect of every type of PA designation at a national scale. The value of biodiversity and natural areas for recreational purposes is often highlighted to support economic investment for conservation (Balmford et al., 2009; Walpole et al., 2001). Given the results of our study, it is likely that the value of green, and blue, spaces that are not special for threatened species and habitats is underestimated. In built areas people are attracted to sites with increased species richness (Figure 5) and there is evidence that the mental health benefits of green spaces increase with biodiversity (Fuller, Irvine, Devine-Wright, Warren, & Gaston, 2007; Luck, Davidson, Boxall, & Smallbone, 2011). In an urban context managing for increased species richness and recreation could be a sustainable approach to meet multiple sustainable development goals (SDG 3,6,8,10,11,12,15; United Nations, 2015).

There are some limitations in the data and methodology we used in this study. While many people take photographs during recreational activities, only some post their photographs online; social media users tend to be under 35, well-educated and earn a higher income than people who do not post on social media (Lo, McKercher, Lo, Cheung, & Law, 2011). Besides demographic biases, not all that is experienced is posted online, which means that we are not capturing all the experiences of wildlife watchers but only those that the social media users considered worthy of sharing. There are also issues with sampling bias related to the type of social media used; in this case Flickr users tend to be more experienced nature recreationists who are nature enthusiasts

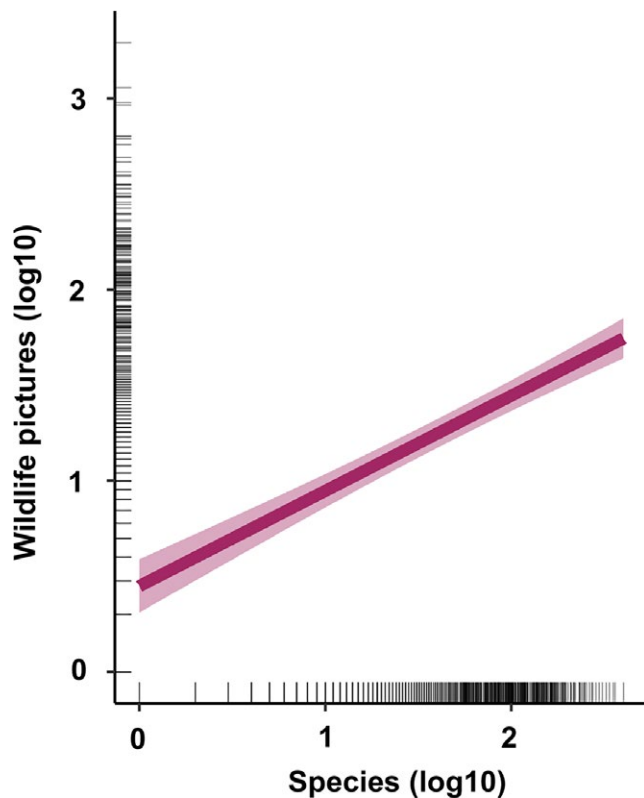


FIGURE 5 Results of species richness model. The plots show predicted values of Flickr users photographing wildlife against the number of species recorded in the same cell while other variables are kept at their mean. Ticks on the axes represent the data, the line represents the predictions from the model and the lighter shaded area is the 95% CIs

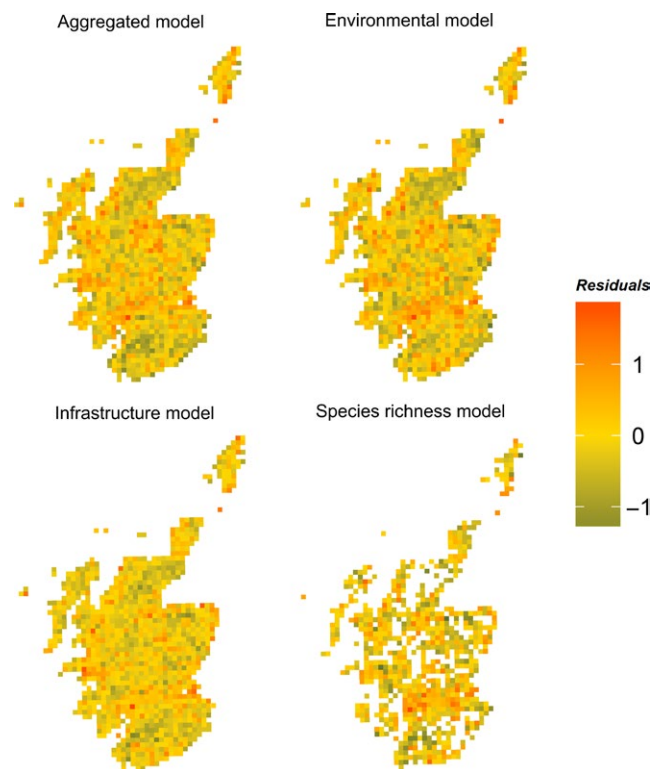


FIGURE 6 Map of model residuals for all the models. Positive residuals (red) indicate that the model is over-predicting, negative residuals (green) indicate under-prediction.

compared to Instagram users who tend to be younger and more interested in charismatic megafauna and other recreational activities (Hausmann, Toivonen, Slotow, et al., 2017). However, the results from a previous validation study (Mancini et al., 2018), together with other recent publications (Hausmann, Toivonen, Slotow, et al., 2017; Levin et al., 2015, 2017; Sessions et al., 2016; Wood et al., 2013), give us enough confidence that the data used in this study is a reliable proxy for wildlife watching activities in Scotland. The choice of wildlife that we focused on could also have affected the spatial patterns of wildlife watching we found. The way in which charisma is defined can influence the effect that charismatic species have on spatial patterns of wildlife watching (Arbieu, Grünewald, Martín-López, Schleuning, & Böhning-Gaese, 2017; Booth, Gaston, Evans, & Armsworth, 2011; Hausmann, Toivonen, Heikinheimo, et al., 2017). However, while seals are one of Scotland's Big Five (<https://scottishwildlifetrust.org.uk/news/can-you-spot-all-of-scotlands-big-5/>) and dolphins are usually considered a charismatic species, the photographs of birds are likely to include species such as golden eagles, also one of Scotland's big Five, but also species such as sparrows and blackbirds not usually considered charismatic. There are also other factors that might be important in influencing the spatial patterns of wildlife watching in Scotland, such as marketing and satisfaction. However, the limited availability of data to quantify these effects limits their inclusion in studies at this scale and resolution. Lastly, there are many more types of human interactions with nature that this study does not include, both recreational, such as outdoor sports or dog walking, and non-recreational, such as spiritual or social-cohesion experiences. Although it is possible that photographs of wildlife were taken while conducting one of these other activities, the data used in this study can only be representative of wildlife watching and, therefore, all the conclusions made are not generalisable to other types of interactions with nature. An interesting extension of this analysis would be to compare total number of Flickr photographs, to wildlife photographs. This would allow us to determine which sites are used primarily for wildlife watching, which sites are used for wildlife watching only secondarily or during other activities and which activities are mostly associated with wildlife recreation.

5 | CONCLUSIONS

The widespread use of the Internet and social media in the nation we targeted allowed us to quantify wildlife watching activities outside PAs where visitor numbers are not usually monitored. Thanks to this wider sampling, we could make inferences about drivers of the attractiveness, and hence the value, of the natural environment outside PAs. We found little overlap between areas of high conservation value and areas of high recreational value (Figures 2–4). This is a positive result for conservation, because it means that those environmental features that are priorities for conservation are to some extent protected from intensive recreational activities. However, this result also questions the synergy between the goals of conservation and those of recreation. Attracting visitors to natural areas requires improving access and building infrastructure, which could

compromise the integrity of the natural features that we want to protect. For example, there are 4,447 species in the IUCN Red List of Threatened Species that are threatened by development of “tourism & recreation areas” (threat 1.3, IUCN, 2018). In managing natural areas, it is important to find a balance between giving people the opportunity to enjoy nature and achieve conservation goals. From our results it seems that designations such as Country Parks and LNR achieve this balance by attracting higher intensities of nature recreational activities and leaving areas of higher conservation value such as MPA or SSSI relatively undisturbed. Moreover, the notion that the general public does not value naturalness as a recreational service, as shown by our results in a wildlife watching context and by Hornigold et al. (2016) in a wider recreational context, is very important for environmental management. The recreational value of natural areas is often used to support public spending for their conservation (Balmford et al., 2009; Wilkie & Carpenter, 1999), but some natural areas, those that are most important for conservation, attract lower numbers of recreationists. If this human-dominated nature is providing most recreational ecosystem services, the general public might be less inclined to support conservation actions to save more remote natural areas that they do not experience directly (Wells & Lekies, 2006). PAs are under increasing pressure to provide financial justification for their existence (Walpole et al., 2001) and tourism is their main tool to generate income. In the attempt to develop PAs into tourism destinations, more infrastructure might be built to attract more people, which could have negative consequences for their conservation objectives.

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AUTHORS' CONTRIBUTIONS

F.M., D.L. and G.M.C. conceived the ideas and designed methodology; F.M. collected and analysed the data and led the writing of the manuscript. All authors contributed critically to the drafts and gave final approval for publication.

DATA ACCESSIBILITY

All the scripts used to collect and analyse the data are available via Zenodo <https://doi.org/10.5281/zenodo.1126604> (Mancini, 2017) and <https://doi.org/10.5281/zenodo.1126605> (Mancini, 2018).

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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